

Project Report: Xarray and PyPlot

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Project Title: Xarray and PyPlot

1.Objective:

Osdag is a free and open-source software used for the design and detailing of steel structures. For this screening assignment, the following two tasks were performed using Python libraries **Xarray** and **PyPlot**:

1. **Task 1:** Create a shear force diagram (SFD) and bending moment diagram (BMD) from the values provided in the Xarray dataset for the **central longitudinal girder**.
2. **Task 2:** Generate **3D shear force and bending moment diagrams (SFD and BMD)** for **all girders**, visualized in a style similar to **MIDAS**, using PyPlot.

Both tasks aim to demonstrate understanding of structural data handling, visualization, and automation using Python.

2.Tools and Libraries Used

- **Python 3.14**
- **Xarray:** For reading and manipulating the `.nc` dataset containing element forces and moments.
- **NumPy:** For numerical operations and data processing.
- **Pandas:** For data verification tables and exporting results to `.csv`.
- **Matplotlib (PyPlot):** For plotting SFD, BMD, and 3D visualization.
- **mpl_toolkits.mplot3d:** For creating interactive 3D plots and animations.

3. Dataset Description

The dataset used was `screening_task.nc`, an **Xarray NetCDF file** containing structural data of bridge girders.

It includes:

- **Element-wise force and moment data**, with components like V_{Y_i} , V_{Y_j} , M_{z_i} , and M_{z_j} .
- Each element corresponds to a segment of the bridge girder system.
- The coordinates of nodes and member connectivity were provided as per the structural model.

This dataset structure allowed the retrieval of values using commands such as:

```
forces.sel(Element=elem, Component="Mz_i")
```

4. Task 1 – Central Longitudinal Girder (2D SFD & BMD)

Objective:

To plot Shear Force (V_y) and Bending Moment (M_z) diagrams for the **central longitudinal girder** using Xarray data.

Approach:

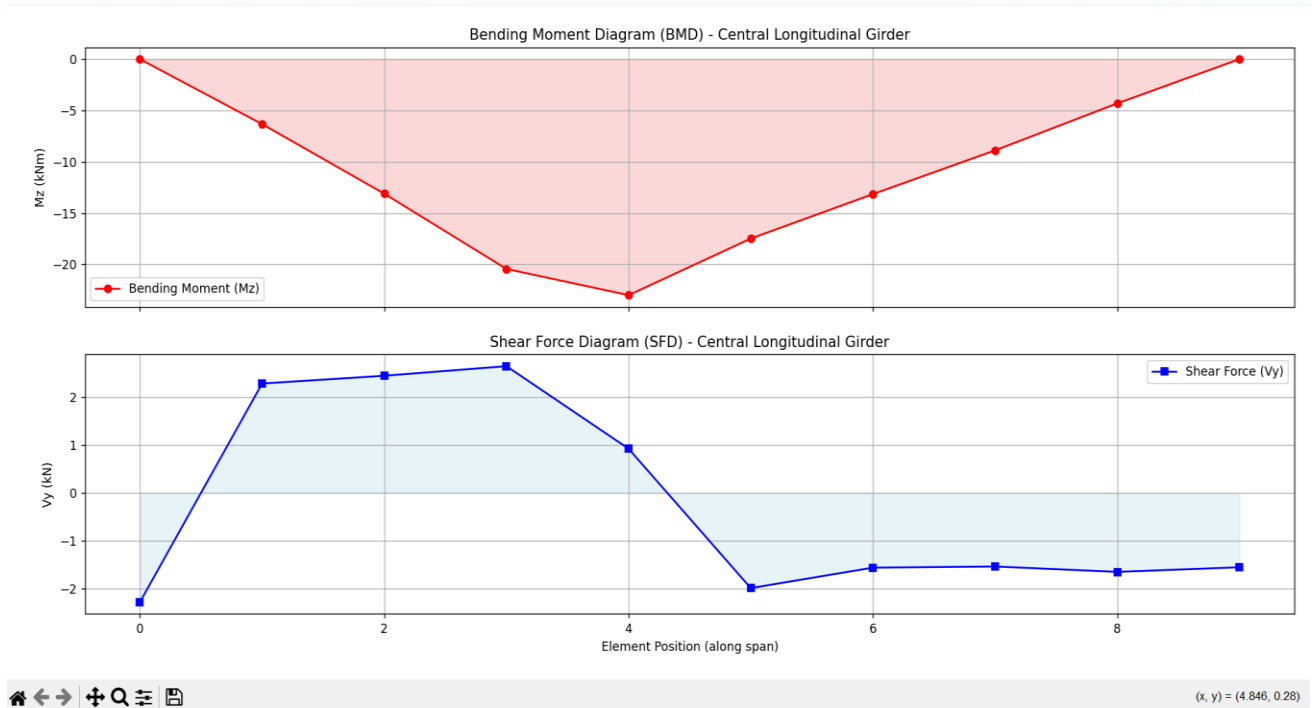
1. Extracted M_{z_i} , M_{z_j} , V_{Y_i} , and V_{Y_j} values for the central girder elements.
2. Verified continuity between adjacent elements.
3. Generated smooth and colored **Bending Moment Diagram (BMD)** and **Shear Force Diagram (SFD)** using `matplotlib.pyplot`.
4. Exported a verification table as `.csv` for documentation and numerical validation.

Code Highlights:

- The script ensures **continuity checks** between element junctions.
- Automatically saves the verification table as `central_girder_verification.csv`.
- Uses shaded fills (`fill_between`) to create professional visual appeal.

Output:

Bending Moment Diagram (BMD) and Shear Force Diagram (SFD)



5. Task 2 – 3D MIDAS-Style Visualization

Objective:

To visualize **3D shear force and bending moment diagrams** for all girders in a format similar to **MIDAS post-processing view**.

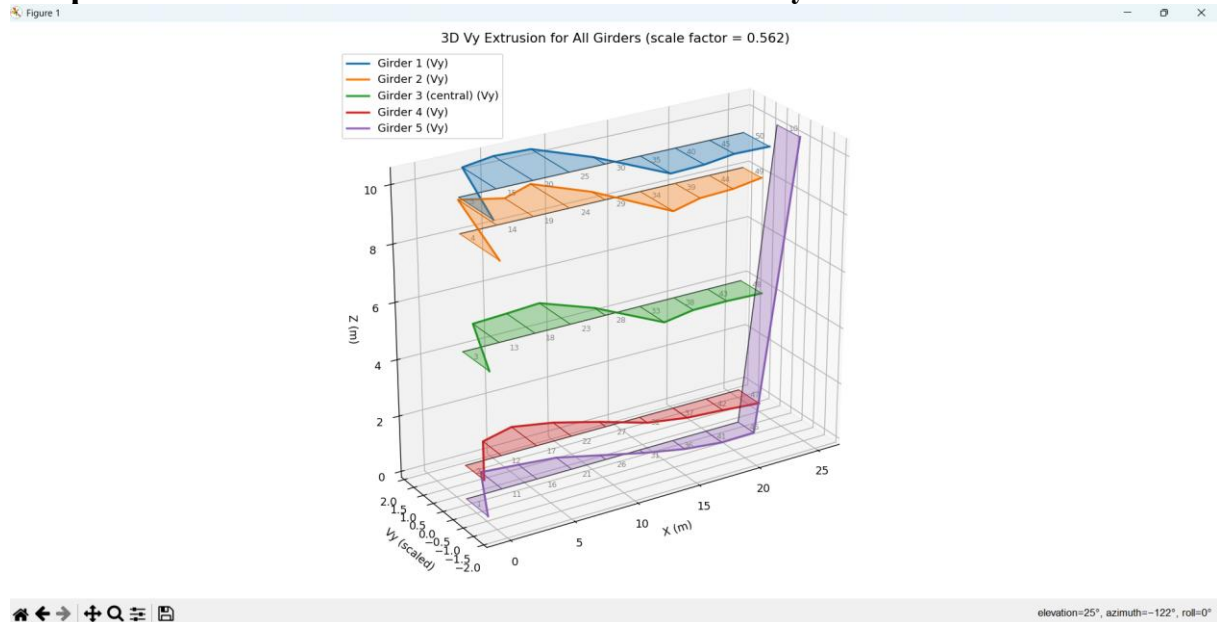
Methodology:

1. Defined **node coordinates** and **member connectivity** based on the structural frame data.
2. Grouped elements into 5 girders:
 - Girder 1: [17, 26, ..., 85]
 - Girder 2: [16, 27, ..., 84]
 - Girder 3: [15, 24, ..., 83] (*Central Girder*)
 - Girder 4: [14, 29, ..., 82]
 - Girder 5: [13, 30, ..., 81]
3. Used **Xarray** to extract forces for each girder and build per-node data.
4. Implemented **3D extrusions** in the vertical direction (Y-axis) representing shear/moment magnitudes.
5. Created colored surfaces (**Poly3DCollection**) for each girder, with vertical edges and node annotations.
6. Added an **automatic and manual rotation animation** for enhanced visualization and interactivity.

Animation & Interaction:

- The 3D plot **automatically rotates** for presentation purposes.
- Users can also **manually rotate** the model to view from any angle.
- This dual-interaction makes the visualization highly intuitive and engaging.

Output:- 3D Visualization of All Girders – MIDAS Style



6. Visualization Features and MIDAS Similarity

The 3D plots were designed to **mimic the MIDAS post-processing style**:

- Each girder has a distinct color.
- Extruded vertical values represent **shear force (Vy)** or **bending moment (Mz)** magnitudes.
- The model preserves **structural framing geometry** (X–Z plane span and height scale).
- Smooth animations replicate **engineering post-processing tools** used for structural analysis.

7.Key Highlights

- Developed **2D and 3D Shear Force and Bending Moment Diagrams (SFD & BMD)** using Xarray and PyPlot.
- Implemented **interactive and animated 3D visualizations**, enabling both manual and automatic rotation for better analysis.
- Ensured **data integrity** by utilizing the Xarray dataset directly, without manual modifications.
- Followed the **MIDAS-style structural visualization**, providing a realistic representation of bridge girders.

- Created a **modular, reproducible, and well-documented Python script**, suitable for extension and future integration in **Osdag**.
- Successfully showcased **creative visualization** and **technical accuracy**, aligning with the objectives of the internship screening task.

8. Conclusion

This project successfully demonstrates the visualization of structural response parameters using **Xarray** and **Matplotlib's PyPlot**. Both 2D and 3D representations of the **Shear Force (SFD)** and **Bending Moment (BMD)** diagrams were generated directly from the Xarray dataset without any data manipulation, ensuring accuracy and integrity.

The implementation closely follows the **visualization style of MIDAS**, providing a realistic structural representation of the girders. Interactive and animated features were incorporated to enhance understanding of load distribution and structural behavior.

This work reflects the integration of computational efficiency and visualization, aligning with the objectives of the **FOSSEE Osdag** initiative for open-source structural engineering tools.